

Foundations of Artificial Intelligence

Introduction: What Is Artificial Intelligence?

Artificial Intelligence, often called **AI**, is a branch of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include things like recognizing speech, understanding images, making decisions, and solving problems.

You already interact with AI more often than you might realize. Examples include voice assistants like Siri or Alexa, recommendation systems on Netflix or Amazon, spam filters in your email, and even navigation apps that suggest the fastest route home.

AI matters because it allows computers to analyze large amounts of information, automate tasks, and support decision-making in ways that would be difficult or impossible for humans to do alone.

Core Ideas Behind Artificial Intelligence

To understand how AI works, it helps to look at a few key ideas that form the foundation of the field.

Intelligent Agents

In AI, many systems are described as **intelligent agents**. An intelligent agent is something that:

- Observes its environment
- Makes decisions based on what it observes
- Takes actions to achieve a goal

For example, a chess-playing program looks at the board, decides on the best move, and then makes that move. A self-driving car observes the road, traffic, and signs, and then decides how to steer, brake, or accelerate.

Data and Knowledge

AI systems depend heavily on **data** and **knowledge**.

- **Data** is information, such as images, text, numbers, or examples of past behavior.
- **Knowledge** is what the system learns or is told about how things relate to each other.

Some AI systems use **rules** that are written by humans. Others use **machine learning**, where the system learns patterns from data instead of being told exactly what to do in every situation.

Learning and Adaptation

One of the most powerful ideas in AI is **learning**. When an AI system learns, it improves its performance over time by studying examples and experiences.

For instance, a movie recommendation system learns what you like by paying attention to what you watch and rate. Over time, its suggestions usually become more accurate because it has learned your preferences.

Reasoning and Decision-Making

AI systems also need to **reason** and **make decisions**. This can involve:

- Following logical rules
- Comparing different options
- Estimating probabilities
- Choosing the best action based on available information

The goal is not just to act, but to act in a way that makes sense for the situation and moves the system closer to its goal.

Types of Artificial Intelligence

Not all AI is the same. There are two major categories you should know about.

Narrow (Weak) AI

Narrow AI is designed to do **one specific task** or a small set of tasks. Examples include:

- Voice assistants
- Face recognition systems
- Spam filters
- Recommendation systems

All AI systems in use today fall into this category.

General (Strong) AI

General AI refers to a system that could perform **any intellectual task** that a human can do. This type of AI would be able to learn, reason, and solve problems across many different areas, just like a person.

At this time, **General AI does not exist**. It is a long-term goal of AI research, not something we currently have.

Why These Foundations Matter

Understanding the foundations of AI helps you:

- Better understand how modern technology works
- Think critically about what AI can and cannot do
- Recognize the limits and risks of AI systems
- Use AI tools more responsibly and effectively

AI is not magic. It is built by people, trained on data, and guided by rules and algorithms. This means it can be powerful, but it can also make mistakes or reflect the biases in its data.

Quick Knowledge Check

Question 1

What is an intelligent agent in AI?

Answer: An intelligent agent is a system that observes its environment, makes decisions, and takes actions to achieve a goal.

Question 2

What is the difference between Narrow AI and General AI?

Answer: Narrow AI is designed to perform a specific task, while General AI would be able to perform any intellectual task a human can. Currently, only Narrow AI exists.

Question 3

Why is data important to AI systems?

Answer: Data is important because it allows AI systems to learn patterns, improve their performance, and make better predictions or decisions over time.

Summary

In this lesson, you learned that Artificial Intelligence is about building systems that can:

- Observe the world
- Learn from data
- Make decisions
- Take actions to achieve goals

These foundations will help you better understand more advanced topics later in the course, such as machine learning, neural networks, and ethical issues in AI.

Slide 1 — Title Slide

Foundations of Artificial Intelligence
CS 600: Artificial Intelligence Foundations
Week 1

Slide 2 — What Is Artificial Intelligence?

- Artificial Intelligence (AI) is a branch of computer science
 - Focuses on creating systems that can perform tasks that normally require human intelligence
 - Examples:
 - Voice assistants (Siri, Alexa)
 - Recommendation systems (Netflix, Amazon)
 - Spam filters and navigation apps
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Slide 3 — Why AI Matters

- Helps analyze large amounts of data
 - Automates tasks
 - Supports decision-making
 - Used in everyday technology, often without us noticing it
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Slide 4 — Core Ideas Behind AI

- AI systems are built around a few key concepts:
 - Intelligent agents
 - Data and knowledge
 - Learning and adaptation
 - Reasoning and decision-making
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Slide 5 — Intelligent Agents

- An intelligent agent:
 - Observes its environment
 - Makes decisions
 - Takes actions to achieve a goal
 - Examples:
 - Self-driving car
 - Chess-playing program
 - Recommendation system
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Slide 6 — Data and Knowledge

- AI systems rely on:
 - Data (examples, information, experiences)
 - Knowledge (rules, relationships, patterns)
 - Two common approaches:
 - Rule-based systems (written by humans)
 - Data-driven systems (machine learning)
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Slide 7 — Learning and Adaptation

- Many AI systems can learn over time
 - Learning means:
 - Improving performance with experience
 - Finding patterns in data
 - Example:
 - Streaming services learning your preferences
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Slide 8 — Reasoning and Decision-Making

- AI systems must choose the best action
 - This may involve:
 - Logic and rules
 - Comparing options
 - Using probabilities
 - Goal: Make the best possible decision with available information
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Slide 9 — Types of Artificial Intelligence

- Two major categories:
 - Narrow (Weak) AI
 - General (Strong) AI
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Slide 10 — Narrow (Weak) AI

- Designed to perform a specific task
 - Examples:
 - Voice assistants
 - Facial recognition
 - Spam filters
 - Recommendation systems
 - All AI in use today is Narrow AI
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Slide 11 — General (Strong) AI

- Would be able to perform any intellectual task a human can
 - Could learn and reason across many areas
 - Does not currently exist
 - Remains a long-term research goal
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Slide 12 — Why These Foundations Matter

- Helps us:
 - Understand how AI really works
 - Recognize its limits
 - Use AI responsibly
 - AI is not magic:
 - Built by humans
 - Trained on data
 - Guided by algorithms
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Slide 13 — Knowledge Check

Let's check your understanding...

Slide 14 — Question 1

What is an intelligent agent?

- A system that:
 - Observes its environment
 - Makes decisions
 - Takes actions to achieve a goal
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Slide 15 — Question 2

What is the difference between Narrow AI and General AI?

- Narrow AI:
 - Performs specific tasks
 - General AI:
 - Would perform any intellectual task a human can
 - Only Narrow AI exists today
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Slide 16 — Question 3

Why is data important to AI systems?

- Data allows AI systems to:
 - Learn patterns
 - Improve performance
 - Make better predictions and decisions
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Slide 17 — Summary

- AI systems can:
 - Observe
 - Learn
 - Reason
 - Act
 - Foundations of AI prepare us for:
 - Machine learning
 - Neural networks
 - Ethical considerations in AI
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Slide 18 — Closing

Next up:

We'll build on these foundations by exploring how machines learn and how data drives modern AI systems.